

# PLUGIN

Project understanding & pitch guide

**An EV charging platform connecting drivers, station operators, and the platform admin.**

## What the project does

PLUGIN brings station discovery, charging reservations, session tracking, wallet payments, invoices, and station administration into one connected system. A driver can plan a charging visit, a station operator can maintain the station and its prices, and an admin can oversee the network and approve new operators.

## The project in one sentence

PLUGIN helps EV drivers plan and manage charging while giving station operators and the platform admin a shared system for running the charging service.

## Who this guide is for

A teammate, business partner, evaluator, presenter, or prospective stakeholder who needs to understand the product without reading its code. It explains the problem, the solution, the three roles, current features, everyday workflows, and a practical way to pitch and demonstrate the project.

## Scope of this edition

Prepared from the local project on 11 September 2026. "Current" means a feature is represented in the reviewed application screens and supporting project logic. This is a product walkthrough based on repository inspection, not a live deployment or hardware certification.

The project contains a customer mobile app, a web experience for customers and management, and a shared central service. Some live services need configuration. Physical charger control and trusted meter-based billing remain integration work.

Plain-language edition | No code walkthrough | Simple, print-friendly layout

# How to use this guide

A complete reading path, with a shorter route for presenters

Read pages 3-4 first to understand the purpose and structure. Pages 5-13 explain how the product works. Pages 14-15 connect the features into one complete journey. Pages 16-18 help you present it accurately.

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## Names used throughout

Customer, user, and EV driver refer to the person charging a vehicle. Station user, station manager, and station operator refer to the approved person managing a station. Admin means the person overseeing the platform as a whole.

For a quick handoff: share the whole PDF, and ask the presenter to focus on pages 3, 4, 14, 17, and 18 before a meeting.

# The problem and the solution

The practical need PLUGIN is designed to address

## The driver problem

Finding a charging station is only the first step. A driver also needs to know where it is, when it operates, which connectors it offers, whether charging capacity is available, what it costs, and whether a visit can be planned around a specific time. After charging, payment and proof of the transaction should be easy to find.

## The station operator problem

A station needs an accurate public listing, a clear record of its charging points, a way to take unavailable equipment out of use, and manageable prices. If these details are disconnected from the customer booking experience, the station and the driver can work from different information.

## The platform admin problem

A growing charging service needs a controlled way to onboard operators, review their documents, monitor reservations and sessions, inspect payments, handle exceptions, and understand activity across stations. Individual station dashboards alone do not provide this network-wide view.

| Problem                          | What PLUGIN provides                                                                              |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| Unclear station information      | Searchable station listings with location, hours, contacts, connectors, status, and prices.       |
| Uncertain timing and capacity    | Scheduled bookings plus mobile arrival-based reservations, with conflict and expiry checks.       |
| Disconnected payment records     | A customer wallet, session-linked billing, payment status, invoices, and history.                 |
| Manual partner onboarding        | A structured station application, document review, approval, and operator access setup.           |
| Fragmented operational oversight | Station management tools and a separate admin view of bookings, customers, sessions, and revenue. |

## The intended outcome

Make charging easier to plan, make station information easier to maintain, and give the platform team a consistent record of what happened. These are the product goals; no measured waiting-time reduction, sales growth, or market adoption is claimed in this guide.

# Product structure and the three roles

One service, three responsibilities, connected records

## The three parts of the product

1. Customer mobile app: the driver experience for stations, navigation, bookings, charging, wallet, history, notifications, vehicles, and account settings. The repository includes an Android project.
2. Web application: public station browsing, customer account and charging workflows, station operator onboarding, and the management workspace used by operators and the admin.
3. Shared central service: keeps accounts, stations, reservations, sessions, wallets, bills, applications, and notifications consistent across the connected interfaces.

| Role                 | Main responsibility                                             | Scope                                                                               |
|----------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Customer / EV driver | Find, book, charge, pay, and manage personal records.           | Own account, vehicles, bookings, sessions, wallet, and bills.                       |
| Station operator     | Maintain approved station details, charging points, and prices. | Assigned stations and a dashboard summarizing their activity.                       |
| Admin                | Approve operators and oversee the entire charging platform.     | Network-wide management, customers, bookings, sessions, revenue, and audit history. |

## How the information connects

Customer + vehicle -> station + charging point -> booking -> charging session -> wallet settlement + bill -> invoice and reports

A station is the physical location. A charging point is an individual connector or charging resource at that location. A booking reserves time; a session records the charging activity. The bill belongs to that session, so the transaction can be traced back to the customer and station.

Station operators share part of the management workspace with the admin, but do not receive the admin-only booking, customer, session, revenue-report, analytics, audit, or application-review pages. The mobile app currently focuses on the customer role.

# Customer: account, vehicles, and discovery

Getting ready to charge and choosing a station

## Account and sign-in

- Create a customer account and confirm registration through an email one-time password (OTP). Resend and recovery flows are included.
- Sign in with account credentials or use Google sign-in when the required Google setup is configured.
- Recover a forgotten password. Profile and mobile settings also provide password-change and account-deactivation flows with identity verification.

## Profile and vehicles

Maintain personal profile details and save vehicles, including identifying details such as make, model, registration, and nickname. A customer can manage multiple vehicles and select an active vehicle. A vehicle must be added before making a charging booking.

Vehicle records linked to bookings cannot simply be removed. This keeps the vehicle history associated with past and current charging activity understandable.

## Station discovery

- Browse station listings and search by location or station information. The web search invites a city, area, or pincode search.
- Inspect station address, coordinates, contact information, operating hours, charging points, connector details, maximum power, and available pricing.
- Use station and charger filters. Mobile includes All, DC Fast, AC, Available, and Favorites filters.
- Save favorite stations on the mobile device and return to them later. Favorites are currently device-local rather than a shared account feature.
- Open a station location and use the mobile navigation experience to plan the journey.

## What the driver should check

Choose a suitable connector using its listed type and power, confirm the station hours, and review the displayed rate before booking. The current mobile AC/DC list filter uses station naming as a shortcut; connector details are the stronger reference for compatibility.

The visible station status reflects the platform record. It should not be described as independently verified live charger telemetry.

# Customer: scheduled bookings and changes

Reserve a specific time, then manage the reservation

## A normal scheduled booking

1. Select a station and charging point, or an available assignment supported by the booking flow.
2. Choose a future date, start time, and charging duration. The current accepted duration is 1 to 60 minutes.
3. Review the charging details and submit. The service checks the station, vehicle, time window, and conflicting reservations before accepting the booking.
4. Open My Bookings or the mobile booking list to see the reservation, timing, station, and current status.

## Rules that make the reservation meaningful

- A customer must complete or cancel an active booking before creating another one.
- An inactive station or an unavailable charging point cannot be treated as bookable equipment. Overlapping reservations for the same charging resource are rejected.
- The chosen scheduled time stays fixed; location estimates do not silently replace it.
- The booking captures the applicable price so a later price edit does not rewrite the agreed rate for that reservation.

## Changing or cancelling a plan

The customer web experience offers a reschedule request with a new time and a reason. The admin can approve or reject it, and the requested time must still pass availability checks. A pending request is not the same as an approved new booking time.

A customer can cancel an eligible booking before charging begins. Cancellation records a reason. Completed bookings and bookings with charging already started cannot be cancelled as if they were unused reservations.

## Understanding booking status

Confirmed means accepted. Modified means an accepted change was made. In progress means charging has begun. Completed means the charging journey finished. Cancelled means the reservation was stopped before use. No show means the applicable arrival or start window expired without charging.

A booking reserves charging time; it does not by itself start electricity flowing. Session start is a separate step.

# Customer: charge on arrival and navigation

A mobile booking option based on estimated arrival time

## Two distinct ways to reserve

The mobile booking flow offers "Choose a time" for a fixed schedule and "Charge on arrival" for an arrival-based reservation. Charge on arrival uses the driver's location to estimate when they can reach the station and reserve a suitable charging window.

## How charge on arrival works

1. The driver grants location access, chooses a station and charging duration, and requests an arrival-based booking.
2. PLUGIN estimates arrival and checks an actual charging resource calendar. It includes a 20-minute late-arrival allowance in the reserved window.
3. If the expected arrival time is full, the response can identify a next available window and expected wait. Requests outside station hours or the arrival-hold rules are rejected.
4. While travelling, fresh location updates can adjust an unlocked reservation. They cannot keep extending the original two-hour arrival hold indefinitely.
5. The physical connector is claimed only when the driver is within one mile of the station and within two minutes of the reserved start. Charging still requires a valid start time and a fresh nearby location.

## Navigation and tracking

The mobile navigation screen provides a map-based station journey with route information. ETA displays distinguish traffic-based estimates from approximate fallback estimates. Background booking tracking can be enabled from booking details when the installed app and location permissions support it.

## Practical limits

Location permission, a working network, and correctly configured map services matter. A force-stopped app, revoked permissions, or operating-system background restrictions can stop updates. The reservation deadline still applies. A refreshed location does not revive an expired booking.

Pitch this as arrival-aware reservation planning. Do not promise an exact arrival prediction, unlimited holds, or guaranteed zero waiting time.

# Customer: the charging session

From a valid reservation to completion and a bill

## Starting charging in the application

The customer opens the booking and starts a session when the booking is eligible. The service checks that the customer owns the booking, the station and connector can be used, the time is valid, and there is no existing session or conflicting connector claim. The wallet must also be ready for charging.

## During the session

- The session screen presents charging progress and timing, with energy and cost information calculated from the current session data.
- The charging point is marked as charging and the booking moves to in progress.
- Wallet monitoring applies charging debits as estimated cost increases. Auto-top-up may replenish the balance when it is enabled and authorized.
- The customer can end the session. The admin also has a session-end action for operational intervention.

## How a session finishes

Sessions can complete at the selected booking end time, through a customer or admin end action, or through wallet-protection handling when funds cannot support continuing. Completion updates the booking, releases the connector when its recorded ownership permits, generates a bill, and attempts wallet settlement.

The resulting invoice and payment status appear in the customer's billing/history journey. Session completion also produces a notification. Invoice email is supported for paid bills when mail delivery is configured.

## What the current energy figures mean

The current service estimates energy from charging-point power, elapsed time, and a configured efficiency factor. This is not a verified reading from a physical charger meter. Cost and energy in a demonstration should be explained as software-calculated figures.

The current charger stop command records a request in application logs. Authenticated charger communication, real start/stop acknowledgements, and trusted meter readings are still required before claiming physical remote control or meter-accurate billing.

# Wallet, payments, invoices, and history

How money and charging records fit together

## Customer wallet

The mobile app includes a dedicated wallet screen with balance, transaction history, manual top-up, withdrawal, and auto-top-up setup. Razorpay supports the payment flow, and the service checks provider payment details before crediting the wallet.

- Manual top-up adds funds through the configured payment provider. The current minimum top-up is Rs. 1,000.
- Auto-top-up requires a verified payment authorization, called a mandate. The default low-balance threshold is Rs. 200 and the default refill amount is Rs. 1,000; supported settings can be changed in the wallet flow.
- Withdrawals refund eligible unused funds to the original top-up payment methods. This is not a general transfer to any bank account or another customer.
- The ledger records credits, charging debits, withdrawals, and other balance adjustments with their status.

## Charging and settlement

The current pricing model is per kilowatt-hour (kWh). A booking preserves its rate. During charging, the wallet is monitored and debited; final settlement accounts for amounts already charged. The resulting bill is marked paid or unpaid.

Illustration only: if the software-calculated energy is 8 kWh and the booking rate is Rs. 20 per kWh, the charging amount is Rs. 160. This is not a fixed PLUGIN tariff. The current bill calculation applies a minimum total of Rs. 1.

## Invoices and outstanding bills

Customers can view bill details, download a PDF invoice, and pay an outstanding bill from their wallet. The web billing screen also exports a CSV statement for a selected date range, limited to 366 days. Mobile provides charging and payment history.

When an unpaid bill remains, the customer experience directs the user to billing/payment until it is settled. The admin can inspect bills, download invoices, and record a manual paid status with an audit trail.

Live payments, mandates, and refunds require provider configuration and operational reconciliation. A successful test payment is not proof that production settlement is ready.

# Station operator: application and approval

How a station joins the platform

## Who can apply

A verified customer account can submit a station manager application through the website. The application supports individuals, proprietorships, partnerships, LLPs, and companies. Required business documents vary with the selected business type.

## What the application collects

| Information group     | Examples of details                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicant identity    | Name, contact details, date of birth, address, government ID, identity document, and selfie.                                                            |
| Business              | Business type and names, PAN, GST where applicable, address, registration details, authorized signatory, and supporting documents.                      |
| Station and property  | Station address and coordinates, owned/rented/leased status, property evidence, electricity connection details, operating hours, and emergency contact. |
| Bank information      | Account holder, bank name, account details, IFSC, and bank proof. Collection does not itself create automated operator payouts.                         |
| Charging installation | Number of chargers, charger and connector types, total capacity, manufacturers, installation photos, and site photos.                                   |

## The approval journey

1. The applicant completes the form and uploads the required evidence. PLUGIN creates an application reference for tracking.
2. The applicant can view their own application and status. Tracking is tied to the signed-in owner rather than being an open public document lookup.
3. The admin reviews the application and documents, records review notes, and approves or rejects it.
4. Approval creates or links the station and updates the approved manager directory. Operator access is configured through the portal access workflow and a one-time password-setup invitation.
5. The operator configures the actual charging points and pricing. The onboarding charger summary does not substitute for individual bookable charging-point records.

The application tracks pending, approved, and rejected states. Rejected applications can be corrected through the permitted resubmission flow; the operator workspace also exposes a Re-KYC entry for applicable updates.

# Station operator: running a station

A focused workspace for the approved operator

## Dashboard for assigned stations

The operator dashboard summarizes activity for the stations assigned to that operator: station and charging-point counts, bookings, active sessions, revenue, calculated energy, and customers associated with those stations. Its scope is different from the admin's network-wide totals.

## Maintain the station listing

- View assigned stations and update their location, contact, and operating information.
- Activate or deactivate an assigned station when the rules allow it. A station with an active charging session cannot be deactivated.
- Keep public-facing information aligned with how the station actually operates.

## Manage charging points

Create and edit charging-point records with their identifier, connector type, fast/slow category, and power. Manage availability or take equipment out of service when appropriate. Reserved and charging states are controlled by the booking/session workflow, rather than being arbitrary status choices.

Removal is restricted where charging-point history must be preserved. This avoids erasing the equipment reference behind a past reservation or session.

## Maintain prices

Create, update, and remove permitted pricing records for assigned stations. The active model is per kWh, associated with the station and applicable charging-point category. Charging-point records must exist before station pricing is configured. Existing booking price snapshots preserve the rate used for those bookings.

## A practical daily routine

1. Check the station dashboard and public listing.
2. Update equipment availability to reflect maintenance or operational changes.
3. Maintain hours, contact information, charging-point details, and prices.
4. Refer booking disputes, reschedule decisions, network-wide reports, and customer account actions to the admin workflow.

Operators can manage assigned stations, charging points, and pricing, and see their dashboard. Creating or deleting stations is admin-only. Dedicated booking administration, customer management, session intervention, and detailed revenue-report pages are also admin-only in the current website.

# Admin: managing the whole platform

Network-wide visibility, approvals, and exception handling

## Network dashboard and analytics

The admin can inspect overall station and charging-point counts, bookings, active sessions, customer totals, revenue, and calculated energy. The analytics page presents summary activity metrics; this should not be pitched as predictive AI analytics.

## Station network and operator onboarding

- Create, edit, activate/deactivate, and remove eligible station records; maintain charging points and pricing across stations.
- Search and filter operator applications, inspect submitted details, download supporting documents, and approve or reject applications with review notes.
- Configure operator portal access and review the approved station manager directory. Removing a station is restricted when booking or billing history exists.

## Customers, bookings, and sessions

- Search customers, filter by active status, and activate or deactivate customer access through the supported controls.
- View bookings and their statuses, inspect station-specific reservations, and cancel eligible bookings with a reason.
- Approve or reject customer reschedule requests after availability is checked.
- View charging sessions and use the admin session-end action when intervention is needed.

## Billing and revenue

Review bills and payment history, filter reports by station and date range, inspect revenue totals, download invoices, and mark a bill paid through an audited administrative action. Revenue figures describe the application's billing records; they are not a complete profit-and-loss or bank-settlement report.

## Operational traceability

Audit logs record significant changes and the actor responsible. The admin can inspect these records and filter by entity type. A notifications page helps the admin review notifications belonging to their account; it is not a customer support ticketing system or a mass-message campaign tool.

The admin is responsible for platform governance: who can operate, what the network offers, how exceptions are handled, and what the records show.

# Notifications, records, and access controls

The supporting features that keep the experience connected

## Notifications and reminders

The product has an in-app notification feed, unread counts, individual read actions, and mark-all-read. Relevant workflows create messages around bookings, charging completion, wallet activity, and application events.

The mobile app includes booking-start device reminders and notification actions that can open relevant booking or payment details. These depend on device permissions and the installed app. Email services support registration and recovery OTPs, application tracking/access setup, and paid invoices when delivery is configured.

## Records kept by the platform

| Record                         | Why it matters                                                                   |
|--------------------------------|----------------------------------------------------------------------------------|
| Account and vehicle            | Identifies the driver and the vehicle associated with charging.                  |
| Station, point, and price      | Describes where charging happens, its capacity, and the rate offered.            |
| Booking and session            | Connects reserved time with the activity that actually occurred in the software. |
| Wallet ledger and bill         | Explains money movement, session charges, payment status, and invoice history.   |
| Application and review         | Records who applied to operate a station and what the admin decided.             |
| Notification and audit history | Keeps users informed and helps explain significant changes.                      |

## Access is tied to responsibility

Customers access their own private records. Station operators access assigned stations and their related management data. Admin-only actions remain separate. Document access and financial actions include ownership checks and auditing where applicable.

## Account safety and lifecycle

Email verification, password recovery, verified password changes, and session invalidation support account control. The "delete account" flow currently deactivates the account and revokes access; it should not be described as immediate erasure of every historical record.

These are implemented controls, not a claim of regulatory certification or complete production security. Public operation also needs the project's documented infrastructure, payment, charger, and data-handling launch work.

# Feature and role quick reference

Use this page when explaining who can do what

| Capability                               | Customer                        | Station operator                | Admin                                     |
|------------------------------------------|---------------------------------|---------------------------------|-------------------------------------------|
| Station discovery and details            | Web and mobile                  | Public listing + own management | Public listing + all management           |
| Profile and vehicles                     | Own account                     | Operator access / application   | Customer access controls                  |
| Scheduled booking                        | Web and mobile                  | Dashboard summary               | Inspect and intervene                     |
| Arrival-based reservation                | Mobile                          | Dashboard summary               | Oversight via booking records             |
| Reschedule request                       | Customer web                    | No dedicated review page        | Approve or reject                         |
| Session start / end                      | Own eligible booking/session    | Dashboard summary               | View sessions and end                     |
| Wallet top-up / auto-top-up / withdrawal | Dedicated mobile wallet         | No operator payout module shown | Financial oversight; not a wallet console |
| Bills and invoices                       | Own bills; wallet payment; PDFs | Dashboard revenue summary       | All bills; PDFs; mark paid                |
| Billing statement export                 | Web CSV by date range           | No dedicated export page        | Revenue and payment reports               |
| Station details and status               | View                            | Assigned stations               | All stations                              |
| Create / delete stations                 | No                              | No                              | Yes, subject to history rules             |
| Charging points and pricing              | View before booking             | Manage assigned stations        | Manage across network                     |
| Operator application                     | Submit on web                   | Own status / permitted Re-KYC   | Review and access setup                   |
| Customer management                      | Own account only                | No                              | Search; activate/deactivate               |
| Analytics and audit pages                | No                              | Assigned-station dashboard only | Network-wide                              |
| Favorites and navigation                 | Mobile                          | No dedicated operator flow      | No dedicated admin flow                   |

Web and mobile share the central service but do not expose every feature in identical screens. In particular, advanced wallet controls and arrival-based booking are mobile features, while application review and administration are web features.

All entries describe the reviewed product interfaces. They do not imply that every integration is enabled in the currently deployed environment.

# A complete example

An illustrative journey connecting all three roles

## 1. A station enters the platform

Meera wants to list her EV charging station. She creates a verified customer account, opens the station manager application, enters identity, business, property, station, installation, and bank information, and uploads the required evidence.

## 2. The admin reviews the application

The admin checks Meera's application and documents, records review notes, and approves it. The station is created or linked, and the operator access workflow lets Meera set up access to the management portal.

## 3. The operator prepares the station

Meera confirms the location and opening hours, adds the individual charging points, identifies each connector and its power, and sets a per-kWh rate. The station can now be discovered with meaningful details and reservable equipment.

## 4. A driver plans a visit

Arjun signs into the customer app, adds his vehicle, searches for a station, and inspects Meera's connectors and price. He chooses a fixed 30-minute slot, or uses Charge on arrival if he wants a location-based reservation. PLUGIN checks capacity before confirming.

## 5. The driver starts and finishes a session

Arjun arrives with wallet funds ready and starts the eligible booking. The application records a charging session, updates point status, and monitors the session cost. When the selected time ends or Arjun stops early, it calculates the final energy and charge, completes the booking, and generates a bill.

## 6. Payment and oversight complete the journey

The wallet settles the charge where funds allow. Arjun sees the payment status and downloads his invoice. Meera's station dashboard reflects the activity. The admin can trace the booking, session, bill, and relevant audit records across the platform.

## What happens when the normal path fails

- Slot conflict: the reservation is rejected or an arrival-based alternative is identified.
- Changed plan: request a reschedule or cancel before charging begins.
- Late arrival or expired hold: the start rules prevent using an expired reservation.
- Insufficient funds: the wallet may attempt authorized top-up; an unpaid balance must be settled through payment.

Names and circumstances are fictional. This example describes the application workflow; live charger control and real payment reconciliation require their corresponding integrations.

# Business value and commercial positioning

How to explain the opportunity without inventing traction

## Value for each stakeholder

| Stakeholder      | Value proposition                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| EV driver        | One place to find a station, plan a charging visit, follow the session, manage payment, and keep invoices.           |
| Station operator | A public digital listing and tools to maintain charging equipment, availability, prices, and station-level activity. |
| Platform team    | A governed network with operator approval, centralized records, administrative intervention, and reporting.          |

## What makes the product story coherent

The strongest pitch is the complete charging journey. Discovery leads into a reservation; the reservation leads into a session; the session leads into payment and an invoice; those same records support station and admin oversight. Mobile arrival-based reservations add a practical travel-planning feature to that core workflow.

## Who could be approached first

Potential early stakeholders include independent charging-station operators and businesses that host EV charging, alongside drivers who need to plan repeat charging visits. These are suggested target segments, not evidence of existing customers, partnerships, or network reach.

## Commercial options to explore

- Operator subscription: charge for access to station-management software and service support.
- Transaction fee or commission: charge an agreed platform fee on successfully completed charging transactions.
- Setup and service fees: charge for onboarding, deployment, integration, or ongoing operational support.

These are proposed business models. The current project demonstrates customer energy pricing and payment records; it does not establish a subscription engine, automatic commission splitting, operator payout system, or a finalized commercial contract.

## Evidence to gather in a pilot

Track reservation-to-session completion, cancellations and no-shows, session success, station utilization, repeat customers, payment/refund outcomes, and operator support effort. These measures would help test the value proposition. They are suggested pilot measures, not claimed current results.

Do not use wallet deposits as a proxy for earned revenue, and do not present charging revenue as profit. Electricity, provider fees, station agreements, operations, and support still affect the business economics.

# Ready-to-use pitch and demonstration

A simple talk track someone can use immediately

## A 30-second pitch

PLUGIN connects EV drivers, station operators, and the platform admin in one charging service. Drivers can find stations, reserve time, manage charging, pay through a wallet, and keep invoices. Operators maintain their station, charging points, and prices, while the admin approves operators and oversees the network. The goal is to make charging easier to plan and easier to manage.

## A two-minute explanation

An EV driver needs more than a station address. They need connector information, opening hours, a suitable time to charge, clear pricing, and a reliable record of payment. Station operators need to maintain those details, and the platform team needs to oversee the whole service.

PLUGIN connects these responsibilities. A customer registers, adds a vehicle, finds a station, and reserves a charging session. On mobile, they can either choose a fixed time or reserve around their estimated arrival. The charging workflow tracks the session and its calculated cost, then creates a bill and invoice linked to wallet payment.

A station operator joins through a structured application and document-review process. After approval and access setup, they manage their station details, charging points, and prices. The admin reviews applications and has the wider tools for customers, bookings, sessions, payments, reporting, and audit history.

The current project implements this connected software workflow. A production rollout still needs authenticated physical charger integration, trusted meter data, and dependable live payment reconciliation. The next practical step is a controlled pilot that proves the full journey with real equipment and operational evidence.

## A clear demonstration order

1. Show a station listing and connector details. Explain the driver's decision.
2. Show a customer vehicle and a booking. Explain fixed time versus mobile arrival-based booking.
3. Show session status, wallet activity, and a resulting invoice.
4. Switch to an operator account and show station, charging-point, and pricing management.
5. Switch to the admin and show application approval, bookings, revenue, and an audit record.

Use prepared test accounts and clearly label test payments and calculated charging data. Close with a concrete pilot proposal, not unverified promises about scale, hardware compatibility, or financial returns.

# Questions a stakeholder may ask

Short, accurate answers for a pitch or handoff

## **Is this only a station-finder app?**

It includes station finding, but also reservations, software charging sessions, wallet payment, invoices, operator onboarding, station management, and admin oversight.

## **Do station operators have the same powers as the admin?**

No. Operators maintain assigned stations, points, and pricing and see a scoped dashboard. The admin handles network-wide records, application review, customer access, booking interventions, sessions, and detailed reporting.

## **Does it control a real charger today?**

The reviewed project has a software session workflow and a logged stop-command placeholder. Real protocol integration, device acknowledgements, and physical-state reconciliation remain required work.

## **Are energy use and charging cost measured from the car?**

Current energy is estimated from configured charger power, elapsed time, and efficiency. Billing uses that calculated energy and the booking rate. Trusted meter readings have not been demonstrated by this review.

## **Are payments implemented?**

Razorpay-backed top-ups, mandates, refunds to original payment methods, wallet debits, and bill settlement are represented in the project. Live operation still needs provider configuration, reliable event handling, reconciliation, and failure-recovery evidence.

## **Can we say it is production-ready or already widely used?**

No such evidence was established here. The repository describes controlled testing and outstanding public-launch work. This guide does not verify a deployed station count, app-store release, active users, partnerships, or operating revenue.

## **Does it include AI, automatic payouts, or every charger network?**

The reviewed features do not establish AI recommendations, universal charger interoperability, automatic operator payouts, loyalty programs, a fleet console, or a support ticketing module. Those can be future product decisions, not current pitch claims.

## **What should happen before a live pilot or rollout?**

Connect and test real charger control and meter data; prove payment, refund, and recurring-payment recovery; configure maps and email; secure private documents and deployment; and test the complete driver/operator/admin journey. The project launch document provides the detailed readiness requirements.

# Glossary and basis of this guide

Useful terms and where the project understanding came from

| Term                     | Plain-language meaning                                                        |
|--------------------------|-------------------------------------------------------------------------------|
| EV                       | Electric vehicle.                                                             |
| Station / charging point | The charging location / an individual charging resource at that location.     |
| kW / kWh                 | Power (rate of energy delivery) / energy (quantity used for charging prices). |
| Booking / session        | A reservation for time / the application record of charging activity.         |
| ETA                      | Estimated time of arrival, based on route or approximate travel information.  |
| KYC                      | Identity and business review for operator onboarding; not a certification.    |
| OTP                      | A one-time password used to verify an account action.                         |
| Mandate                  | A customer authorization for supported recurring or automatic payment.        |
| Wallet ledger            | The history explaining credits and debits to a customer wallet.               |
| Audit log                | A record of a significant action, who performed it, and its context.          |
| OCPP                     | A charger communication protocol. Real integration remains pending.           |

## Review basis

The review covered project and launch documentation, web and mobile screens, role permissions, and supporting booking, charging, wallet, billing, onboarding, profile, notification, and reporting workflows.

Source areas: README.md; docs/SECURITY-HARDENING.md; plugin-frontend/src; plugin\_app/App.js and plugin\_app/src; plugin-backend/src/main/java/com/plugin. These references help future maintainers update the guide.

## What was not verified

No live environment, private customer database, real payment, physical charging device, market statistic, or financial performance was tested for this document. No secret configuration values are included. Proposed commercial options and pilot measures are explicitly suggestions.

Edition: 11 September 2026. Recheck the role matrix, booking rules, wallet limits, integrations, and launch status when the project changes.